Analysis and Design of the Mechanism of Action (MoA) Prediction System

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1. Introduction

Mechanism of Action (MoA) prediction is a highly non-linear problem. Small changes in critical metadata, such as exposure time (`cp_time`), can amplify model error, creating a "Black Box" risk in biomedical AI.

This work develops a robust system supported by **Stability Principles** and **Chaos Theory** to ensure reproducibility.

2. Objectives

Problem: Biological data is inherently sensitive to structural noise and input bias.

Goal: Design a modular architecture capable of:

- Detecting data instabilities.
- Quantifying uncertainty.
- Ensuring calibrated predictions under perturbed conditions.

3. Proposed Solution

The architecture integrates three pillars to reduce chaotic sensitivity:

1. **Adversarial Validation:** Detects input distribution shifts before inference.
2. **Mixture of Experts (MoE):** Specialized sub-models handle dataset heterogeneity.
3. **Bayesian Uncertainty:** Quantifies the reliability of each prediction.

4. Experimental Setup

System stability was evaluated through two complementary scenarios:

Scenario 1 (Data-Driven): Training cycles comparing *Baseline* (clean) vs. *Perturbed* conditions (increasing weight of `cp_time=24h` by 15%).

Scenario 2 (Event-Based): Cellular Automaton simulation on a 100×100 grid varying the reactivity parameter R to find bifurcation points.

5. System Architecture

The system is decoupled into functional layers for Loading, Preprocessing, Training, and Reporting.

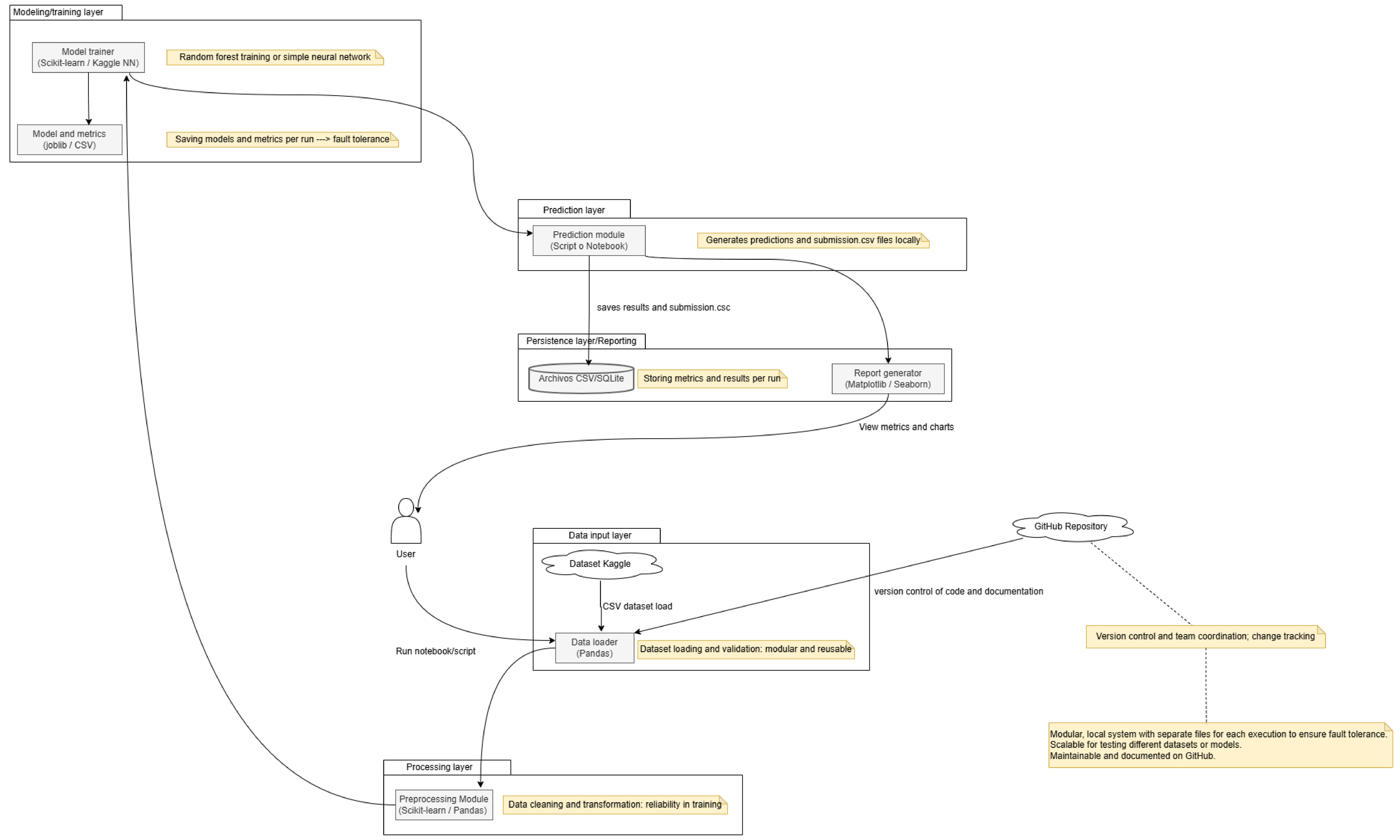

Figure 1:Modular Architecture Diagram illustrating the fault-tolerant pipeline.

Sc. 2: Dynamics Analysis

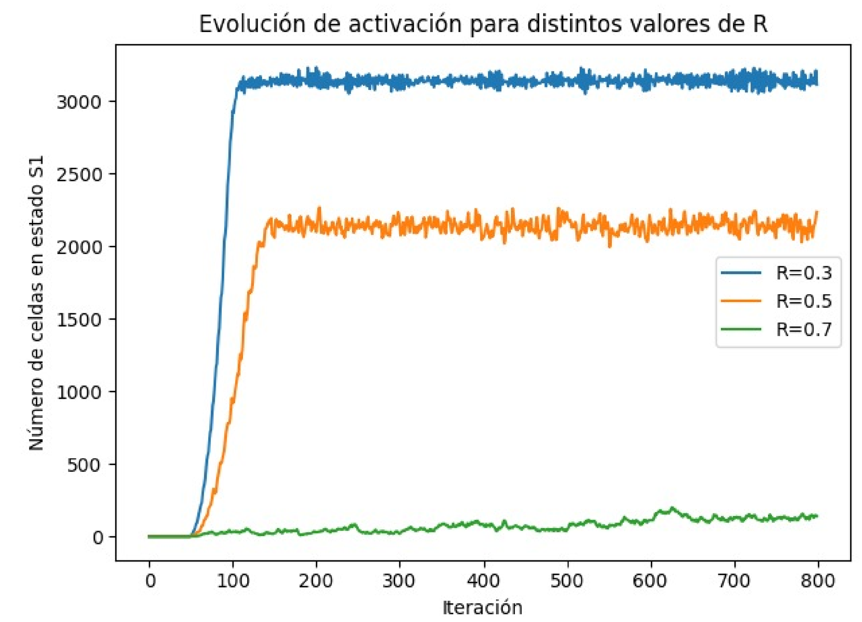

Figure 2:**Temporal Evolution:** Signal instability observed at $R = 0.7$.

Sc. 2: Spatial Patterns

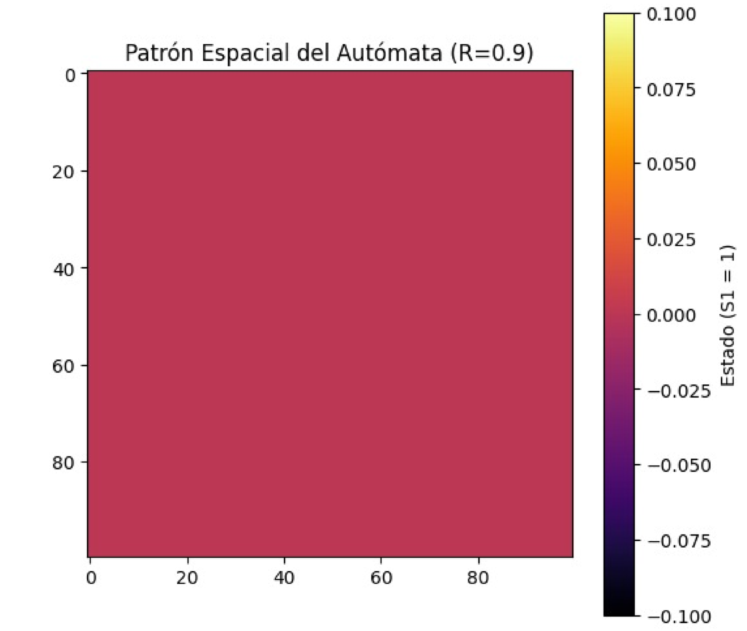

Figure 3:**Chaotic Regime** ($R = 0.9$)**:** System collapse into fractal patterns.

6. Sensitivity Results (Scenario 1)

Impact of temporal bias on predictive performance:

Condition	Log Loss	Effect
Baseline	17.58	-
Perturbed	17.75	**+1% Error**

Table 1:Degradation due to metadata bias.

This confirms that the system is structurally vulnerable to non-random metadata changes.

Critical Finding

Simulations confirm that for reactivity values $R > 0.65$, the system undergoes a **bifurcation**, transitioning from a stable regime to a chaotic state.

7. Conclusions

- **Vulnerability:** The MoA prediction system is sensitive to temporal biases in data.
- **Chaos:** Bifurcation points exist where prediction becomes unreliable without safeguards.
- **Validation:** The proposed modular architecture successfully isolates these instabilities.

References

[1] Laboratory for Innovation Science at Harvard, 'MoA Prediction,' Kaggle, 2020.

[2] S. H. Strogatz, *Nonlinear Dynamics and Chaos*, Westview Press, 2014.

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